

Lecture 09-16

Strogatz Section 6.5: Conservative Systems

→ Conservative systems are subject to a constraint on the transfer of energy (i.e., it is conserved)

→ Newton: $M\ddot{x}_i = F(x_i)$ no t dependence, no friction
 $\Rightarrow$ conservative

→ "when you push, it pushes back" → sign convention

$$F = -\frac{\partial V}{\partial x}$$
$$\vec{F} = -\nabla V$$

V : one function

(Hamiltonian)
evolves system forward
via a conservative law

$$M\ddot{x} + \frac{\partial V}{\partial x} = 0$$
$$\Rightarrow M\dot{x} \cdot \ddot{x} + \frac{\partial V}{\partial x} \cdot \dot{x} = 0 \Rightarrow \frac{d}{dt} \left(\frac{1}{2} M \dot{x}^2 + V \right) = 0 \equiv \frac{d}{dt} E$$

$\Rightarrow$

→ conservative systems were the state of the art until 20th century

→ State space is filled with trajectories, like a ball of spaghetti, each trajectory folds through others, but none intersect

→ these are fibers, which originate from a base manifold; this base manifold specifies the fiber structure

→ More abstractly, these are foliated sheets v. symplectic geometry

6.5.2 Double Well

→ take a potential $V(x) = \frac{1}{2}x^2$ (Hookean)

→ now, consider $V(x) = -\frac{1}{2}x^2$ → direction of acceleration is reversed

→ $V(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2$
need even powers in potential, to keep it from diverging

$$\Rightarrow V(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2$$

System

$$\dot{x} = y$$

$$\dot{y} = x - x^3$$

} → Equilibria automatically require $y = 0 = \dot{x}$
→ one possible, trivial solution is $y = x = 0$
→ $x(1 - x^2) \Rightarrow (x = 1, y = 0)$

→ Now, consider the stability of the system @ these equilibria: (Stability matrix $\underline{A}$)

$$\underline{A}(x, y) = \begin{pmatrix} 0 & 1 \\ 1-3x^2 & 0 \end{pmatrix} = \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix}$$

$$\underline{A}(0, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \underline{A}(\pm 1, 0) = \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix}$$

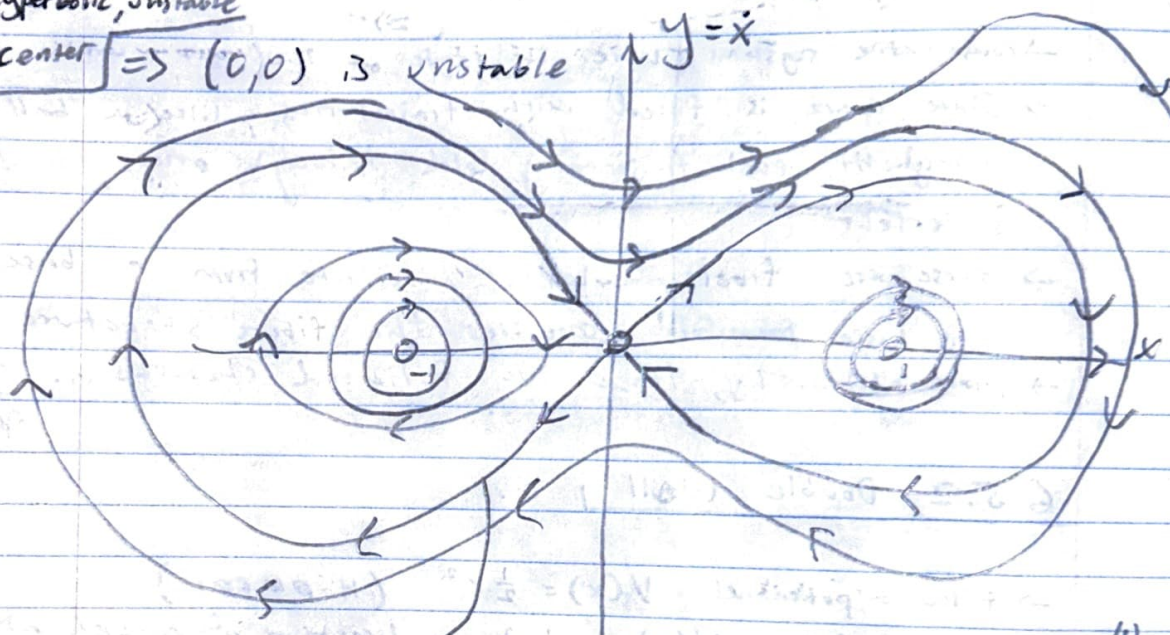
$$\rightarrow \text{Tr } \underline{A} = 0, \Delta_{\underline{A}} = 3x^2 - 1$$

$$\lambda_{(0,0)} = \pm 1$$

$$\lambda_{(\pm 1, 0)} = 0 \pm i\omega()$$

$(0, 0) \rightarrow$ hyperbolic, unstable

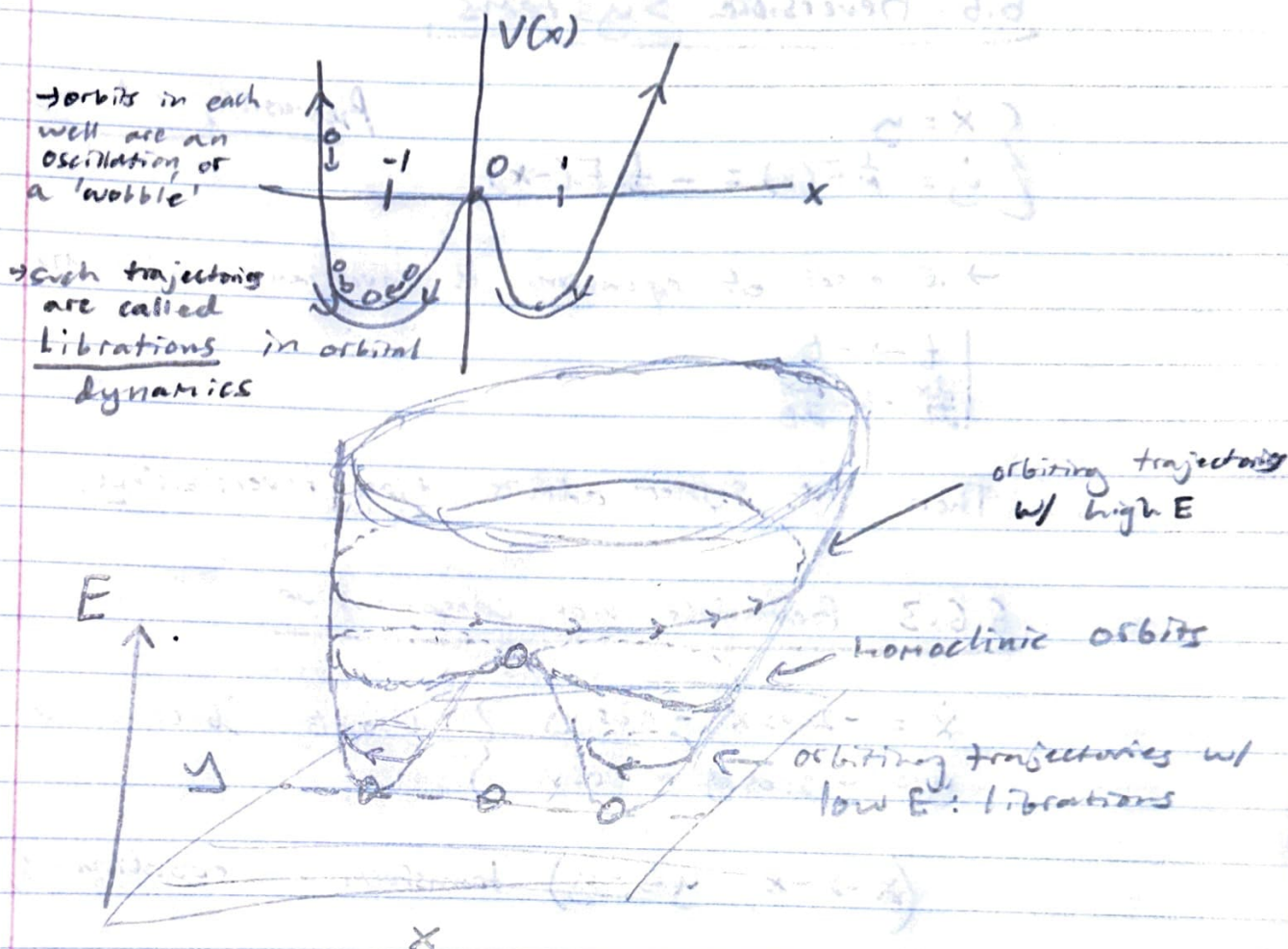
$(\pm 1, 0) \rightarrow$ center $\Rightarrow (0, 0)$ is unstable



→ These trajectories that return to their "coordinates" in phase space at a later time, are called homoclinic orbits: repeatedly orbit through equilibria

→ This double-well potential is actually integrable. Since the trajectories are closed.

→ now we plot the potential: $V(x) = \frac{x^4}{4} - \frac{x^2}{2}$



→ Energy shells elucidate the trajectories' behavior

$$\ddot{x} = \frac{1}{2}x^2 + \frac{1}{4}x^4$$

$$\begin{aligned} t &\rightarrow -t \\ \frac{d}{dt} &\rightarrow -\frac{d}{dt} \\ \frac{d^2}{dt^2} &\rightarrow \frac{d^2}{dt^2} \end{aligned}$$

→ Entropy: even though mechanics is microscopic, only time-reversible if its described with an infinite degree of precision

"Chaos explains thermodynamics in modern understanding"

6.6: Reversible Systems,

Reversible, and conservative

$$\begin{cases} \dot{x} = y \\ \dot{y} = \frac{1}{m} F(x) = -\frac{1}{m} F(-x) \end{cases}$$

→ if a set of equations is invariant under the transform

$$\begin{cases} t \rightarrow -t \\ \frac{dx}{dt} \rightarrow -\frac{dx}{dt} \end{cases}$$

Then the system admits time-reversibility.

6.6.3 Reversible, not conservative

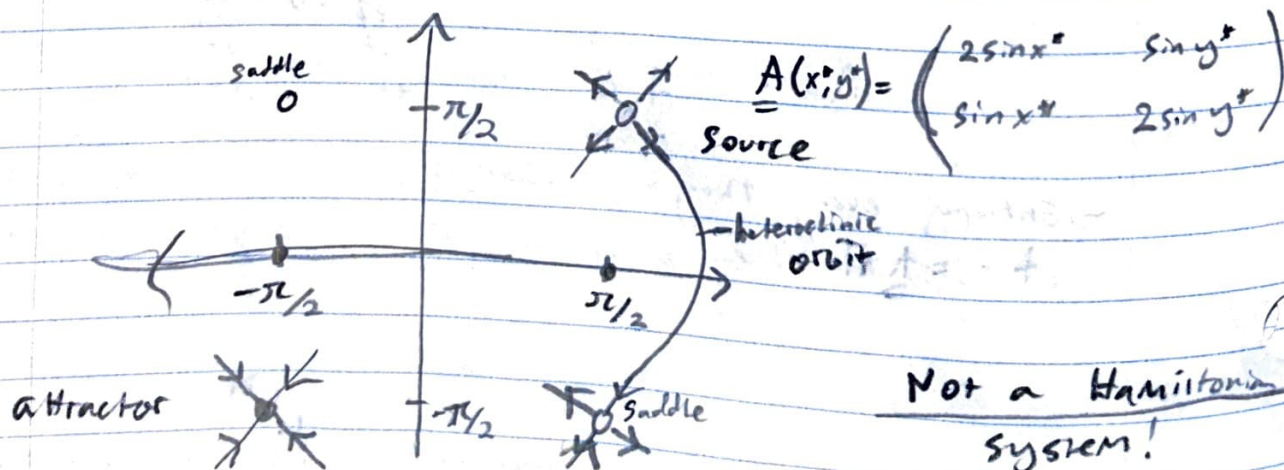
$$\begin{cases} \dot{x} = -2\cos x - \cos y \\ \dot{y} = -2\cos y - \cos x \end{cases} \quad \left. \begin{array}{l} t \rightarrow -t \end{array} \right\} \text{breaks the system}$$

$(x \rightarrow -x, y \rightarrow -y)$ transform of coordinates, 180° rot

$\Rightarrow$ recover original system.

→ it is time reversible, but only under coord. transform

→ Equilibria are $(\pm\pi/2, \pm\pi/2)$:



Not a Hamiltonian system!